Data Analysis
AMS 315

Chapter 12 (Part 2)

Chi-Square Tests and Nonparametric Tests

Learning Objectives

In this Part 2 of Chapter 12, you learn:

- How to use the chi-square test to evaluate the goodness of fit of a set of data to a specific probability distribution
- How and when to use other nonparametric tests (Wilcoxon Rank-Sum and Kruskal-Wallis tests)

Chi-Square Goodness-of-Fit Test

- Does sample data conform to a hypothesized distribution?
  - Examples:
    - Are technical support calls equal across all days of the week? (i.e., do calls follow a uniform distribution?)
    - Do measurements from a production process follow a normal distribution?

Chi-Square Goodness-of-Fit Test

(continued)

- Are technical support calls equal across all days of the week? (i.e., do calls follow a uniform distribution?)
  - Sample data for 10 days per day of week:

	Sum of calls for this day:
Monday	290
Tuesday	250
Wednesday	238
Thursday	257
Friday	265
Saturday	230
Sunday	192
	$\Sigma = 1722$

Logic of Goodness-of-Fit Test

- If calls **are** uniformly distributed, the 1722 calls would be expected to be equally divided across the 7 days:

$$\frac{1722}{7} = 246 \quad \text{expected calls per day if uniform}$$

- Chi-Square Goodness-of-Fit Test: test to see if the sample results are consistent with the expected results

Observed vs. Expected Frequencies

	Observed f_o	Expected f_e
Monday	290	246
Tuesday	250	246
Wednesday	238	246
Thursday	257	246
Friday	265	246
Saturday	230	246
Sunday	192	246
TOTAL	1722	1722

Chi-Square Test Statistic

H_0: The distribution of calls is uniform over days of the week

H_1: The distribution of calls is not uniform

- The test statistic is

$$\chi^2 = \sum_k \frac{(f_o - f_e)^2}{f_e} \qquad (\text{where } df = k - p - 1)$$

where:

k = number of categories

f_o = observed frequency

f_e = expected frequency

p = number of parameters estimated from the data

The Rejection Region

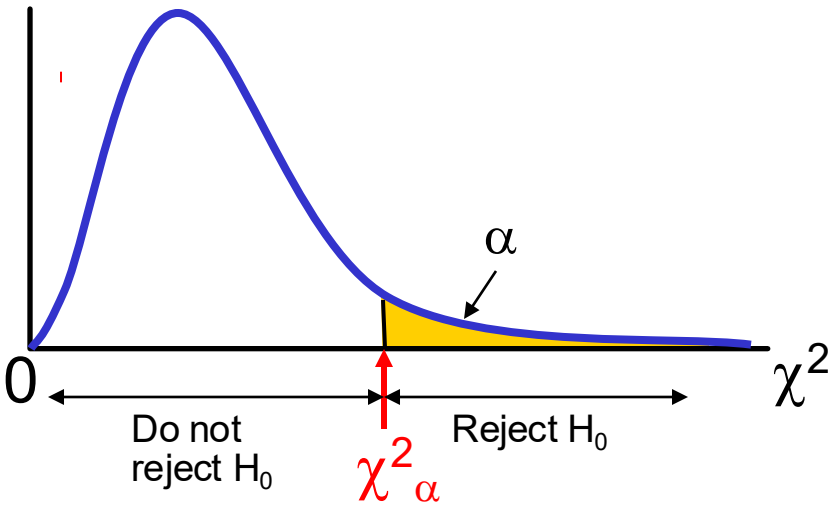
H_0: The distribution of calls is uniform over days of the week

H_1: The distribution of calls is not uniform

$$\chi^2 = \sum_k \frac{(f_o - f_e)^2}{f_e}$$

- Reject H_0 if $\chi^2 > \chi^2_\alpha$

k – 2 degrees of freedom, since p = 1 here (the mean was estimated)



Chi-Square Test Statistic

H_0: The distribution of calls is uniform over days of the week

H_1: The distribution of calls is not uniform

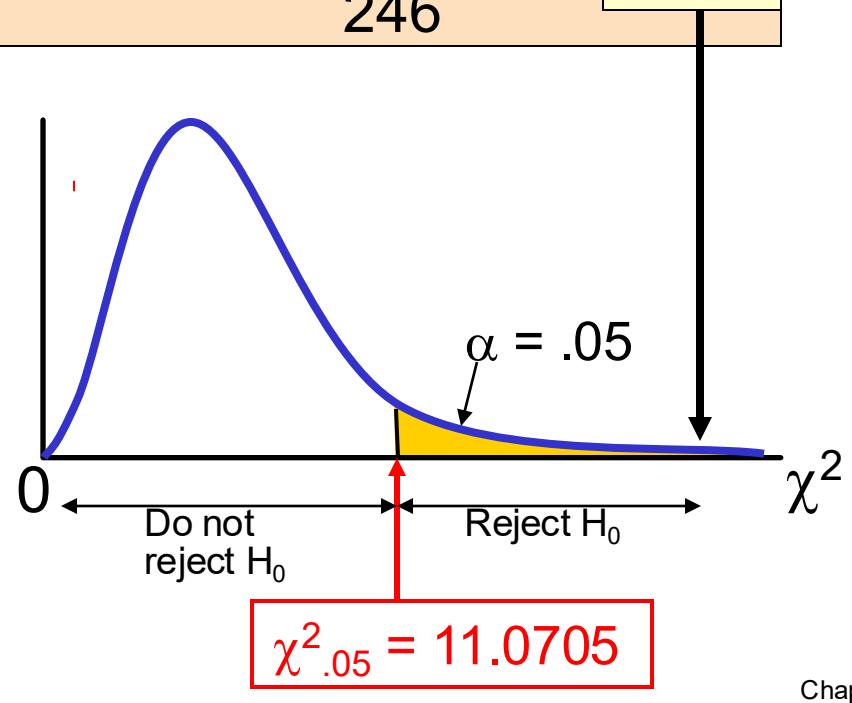
$$\chi^2 = \frac{(290-246)^2}{246} + \frac{(250-246)^2}{246} + \ldots + \frac{(192-246)^2}{246} = \boxed{}$$

k – 2 = 5 (k = 7 days of the week) so use 5 degrees of freedom:

$\chi^2_{.05} = 11.0705$

Conclusion:

χ^2 = _____ > χ^2_α = 11.0705 so **reject H_0** and conclude that the distribution is not uniform



Normal Distribution Example

- Do measurements from a production process follow a normal distribution with $\mu = 50$ and $\sigma = 15$?
- Process:
  - Get sample data
  - Group sample results into classes (cells) (Expected cell frequency must be at least 5 for each cell)
  - Compare actual cell frequencies with expected cell frequencies

Normal Distribution Example

(continued)

- Sample data and values grouped into classes:

150 Sample Measurements
80
65
36
66
50
38
57
77
59
…etc…

Class	Frequency
less than 30	10
30 but < 40	21
40 but < 50	33
50 but < 60	41
60 but < 70	26
70 but < 80	10
80 but < 90	7
90 or over	2
TOTAL	150

Normal Distribution Example

(continued)

- What are the expected frequencies for these classes for a normal distribution with $\mu = 50$ and $\sigma = 15$?

Class	Frequency	Expected Frequency
less than 30	10	?
30 but < 40	21	
40 but < 50	33	
50 but < 60	41	
60 but < 70	26	
70 but < 80	10	
80 but < 90	7	
90 or over	2	
TOTAL	150	

Expected Frequencies

Value	P(X < value)	Expected frequency
less than 30	0.0918	13.77
30 but < 40		
40 but < 50		
50 but < 60		
60 but < 70		
70 but < 80		
80 but < 90		
90 or over		
TOTAL	1.00000	150.00

Expected frequencies in a sample of size n=150, from a normal distribution with μ=50, σ=15

Example:

$$P(x<30) = P\left(Z < \frac{30-50}{15}\right)$$

$$= P(z < -1.33)$$

$$= 0.0918$$

$$(0.0918)(150)=13.77$$

The Test Statistic

Class	Frequency (observed, f_o)	Expected Frequency, f_e
less than 30	10	
30 but < 40	21	
40 but < 50	33	
50 but < 60	41	
60 but < 70	26	
70 but < 80	10	
80 or over	9	
TOTAL	150	150.00

Combine class groups so no class has expected frequency <1

The test statistic is

$$\chi^2 = \sum_k \frac{(f_o - f_e)^2}{f_e}$$

- Reject H_0 if $\chi^2 > \chi^2_\alpha$

(with $k - p - 1$ degrees of freedom)

The Rejection Region

H_0: The distribution of values is normal with $\mu = 50$ and $\sigma = 15$

H_1: The distribution of calls does not have this distribution

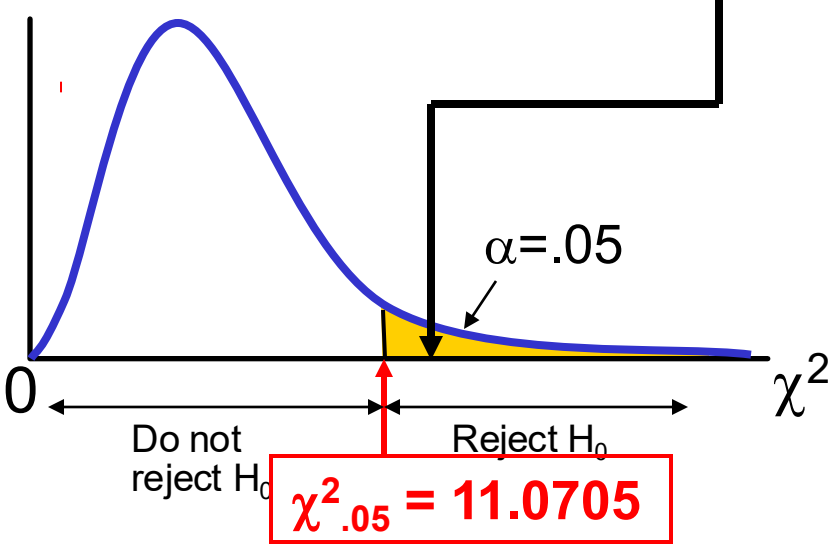
$$\chi^2 = \sum_k \frac{(f_o - f_e)^2}{f_e} = \frac{(10-13.68)^2}{13.68} + \ldots + \frac{(9-3.41)^2}{3.41} = \boxed{}$$

8 classes so use 8 – 2 – 1 = 5 d.f.:

$\chi^2_{.05} = 11.0705$

Conclusion:

Since χ^2 = _____ > χ^2_α we reject H_0 and conclude that there is sufficient evidence that the distribution of calls is not normal.



Wilcoxon Rank-Sum Test for Differences in 2 Medians

- Test two independent population medians
- Populations need not be normally distributed
- Distribution free procedure
- Used when only rank data are available
- Must use normal approximation if either of the sample sizes is larger than 10

Wilcoxon Rank-Sum Test: Small Samples

- Can use when both n_1 , n_2 ≤ 10
- Assign ranks to the combined $n_1 + n_2$ sample observations
  - If unequal sample sizes, let n_1 refer to smaller-sized sample
  - Smallest value rank = 1, largest value rank = $n_1 + n_2$
  - Assign average rank for ties
- Sum the ranks for each sample: T_1 and T_2
- Obtain test statistic, T_1 (from smaller sample)

Checking the Rankings

- The sum of the rankings must satisfy the formula below
- Can use this to verify the sums T_1 and T_2

$$T_1 + T_2 = \frac{n(n+1)}{2}$$

where $n = n_1 + n_2$

Wilcoxon Rank-Sum Test: Hypothesis and Decision Rule

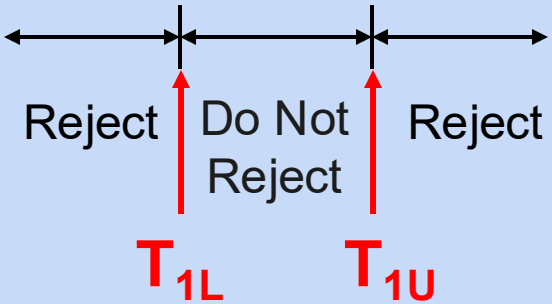
M_1 = median of population 1; M_2 = median of population 2

Test statistic = T_1 (Sum of ranks from smaller sample)

Two-Tail Test

$H_0: M_1 = M_2$

$H_1: M_1 \neq M_2$

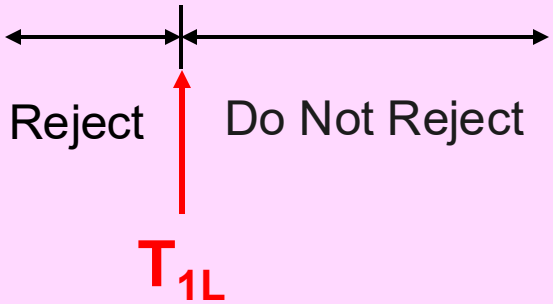


Reject H_0 if $T_1 < T_{1L}$ or if $T_1 > T_{1U}$

Left-Tail Test

$H_0: M_1 \geq M_2$

$H_1: M_1 < M_2$

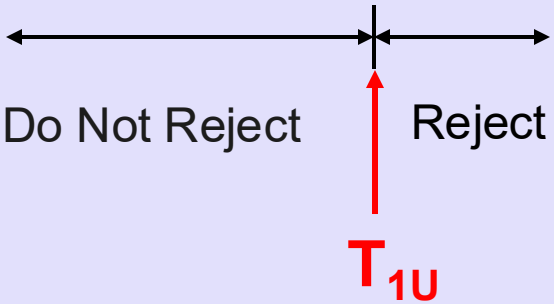


Reject H_0 if $T_1 < T_{1L}$

Right-Tail Test

$H_0: M_1 \leq M_2$

$H_1: M_1 > M_2$



Reject H_0 if $T_1 > T_{1U}$

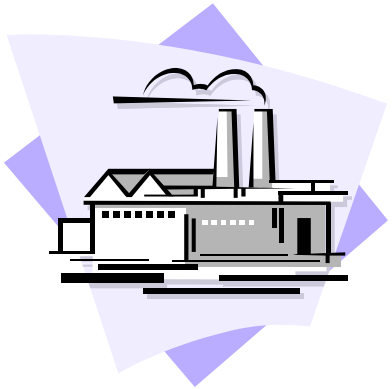
Wilcoxon Rank-Sum Test: Small Sample Example

Sample data are collected on the capacity rates (% of capacity) for two factories.

Are the median operating rates for two factories the same?

- For factory A, the rates are 71, 82, 77, 94, 88
- For factory B, the rates are 85, 82, 92, 97

Test for equality of the population medians at the 0.05 significance level



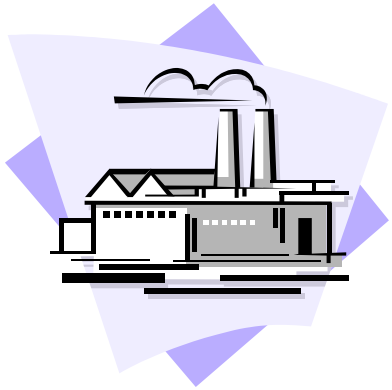
Wilcoxon Rank-Sum Test: Small Sample Example

(continued)

Ranked Capacity values:

Tie in 3rd and 4th places

Capacity		Rank	
Factory A	Factory B	Factory A	Factory B
71		1	
77		2	
82		3.5	
	82		3.5
	85		5
88		6	
	92		7
94		8	
	97		9
	Rank Sums:	20.5	24.5

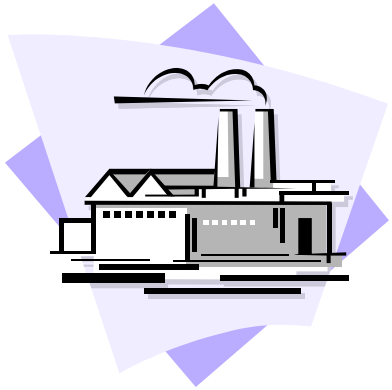


Wilcoxon Rank-Sum Test: Small Sample Example

(continued)

Factory B has the smaller sample size, so the test statistic is the sum of the Factory B ranks:

$$T_1 = 24.5$$



The sample sizes are:

$n_1 = 4$ (factory B)

$n_2 = 5$ (factory A)

The level of significance is $\alpha = .05$

Wilcoxon Rank-Sum Test: Small Sample Example

(continued)

- Lower and Upper Critical Values for T_1 from Appendix table E.6:

n_2	α		n_1	
	One-Tailed	Two-Tailed	4	5
4				
5	.05	.10	12, 28	19, 36
	.025	.05	11, 29	17, 38
	.01	.02	10, 30	16, 39
	.005	.01	--, --	15, 40
6				

$T_{1L} = 11$ and $T_{1U} = 29$

Wilcoxon Rank-Sum Test: Small Sample Solution

(continued)

- $\alpha = .05$
- $n_1 = 4$, $n_2 = 5$

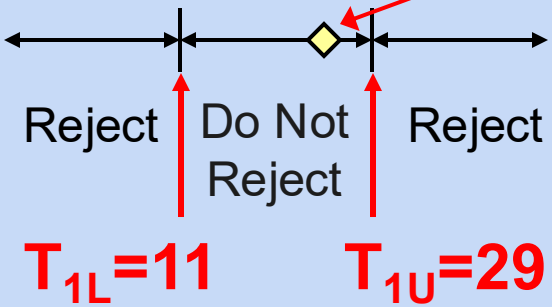
Test Statistic (Sum of ranks from smaller sample):

$T_1 = 24.5$

Two-Tail Test

$H_0: M_1 = M_2$

$H_1: M_1 \neq M_2$



Reject H_0 if $T_1 < T_{1L}=11$

or if $T_1 > T_{1U}=29$

Since $T_{1L}<T_1=24.5<T_{1u}$, we do not reject H_0 and conclude that we have insufficient evidence that the medians are different.

Kruskal-Wallis Rank Test

- Tests the equality of more than 2 population medians
- Use when the normality assumption for one-way ANOVA is violated
- Assumptions:
  - The samples are random and independent
  - Variables have a continuous distribution
  - The data can be ranked
  - Populations have the same variability
  - Populations have the same shape

Kruskal-Wallis Test Procedure

- Obtain relative rankings for each value
  - In event of tie, each of the tied values gets the average rank
- Sum the rankings for data from each of the c groups
  - Compute the H test statistic

Kruskal-Wallis Test Procedure

(continued)

- The Kruskal-Wallis H-test statistic:

  (with c – 1 degrees of freedom)

$$H = \left[\frac{12}{n(n+1)} \sum_{j=1}^{c} \frac{T_j^2}{n_j} \right] - 3(n+1)$$

where:

n = sum of sample sizes in all samples

c = Number of samples

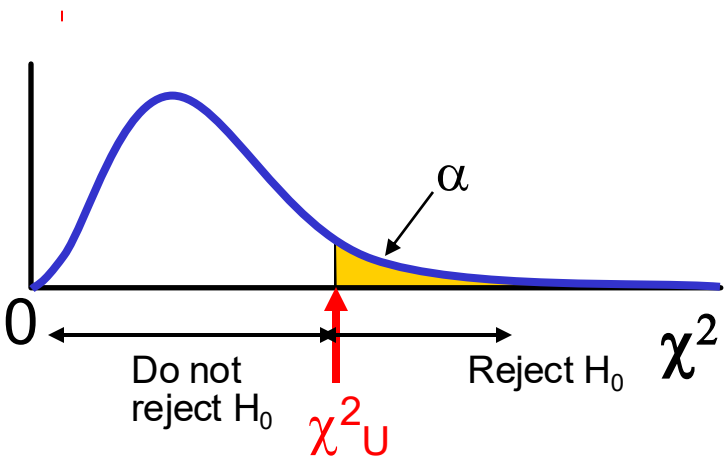
T_j = Sum of ranks in the jth sample

n_j = Number of values in the jth sample (j = 1, 2, … , c)

Kruskal-Wallis Test Procedure

(continued)

- Complete the test by comparing the calculated H value to a critical χ^2 value from the chi-square distribution with c – 1 degrees of freedom



- Decision rule
  - Reject H_0 if test statistic H > χ^2_U
  - Otherwise do not reject H_0

Kruskal-Wallis Example

- Do different departments have different class sizes?

Class size (Math, M)	Class size (English, E)	Class size (Biology, B)
23	55	30
45	60	40
54	72	18
78	45	34
66	70	44

Kruskal-Wallis Example

(continued)

- Do different departments have different class sizes?

Class size (Math, M)	Ranking	Class size (English, E)	Ranking	Class size (Biology, B)	Ranking
23	2	55	10	30	3
41	6	60	11	40	5
54	9	72	14	18	1
78	15	45	8	34	4
66	12	70	13	44	7
	$\Sigma =$		$\Sigma =$		$\Sigma =$

Kruskal-Wallis Example

(continued)

$$H_0: \text{Median}_M = \text{Median}_E = \text{Median}_B$$

$$H_1: \text{Not all population Medians are equal}$$

- The H statistic is

$$H = \left[\frac{12}{n(n+1)}\sum_{j=1}^{c}\frac{T_j^2}{n_j}\right] - 3(n+1)$$

$$= \left[\frac{12}{15(15+1)}\left(\frac{44^2}{5} + \frac{56^2}{5} + \frac{20^2}{5}\right)\right] - 3(15+1) =$$

Kruskal-Wallis Example

(continued)

- Compare H = 6.72 to the critical value from the chi-square distribution for 3 – 1 = 2 degrees of freedom and α = 0.05:

$$\chi^2_U = 5.991$$

Since H = _____ > _____, we reject H_0

and conclude that there is sufficient evidence that at least one population median is not equal.

Chapter Summary

- Applied the χ^2 test for goodness-of-fit test
- Used the Wilcoxon rank sum test for two population medians
- Applied the Kruskal-Wallis H-test for multiple population medians